CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

1.3 Errors & Uncertainties

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Structured Questions (Paper 2) **Total:** 12 Questions (146 Marks)

Question 1 Easy [12 marks]

Part (a) [2 marks]

A physics teacher tells a class:

"Measurements may give a precise value for the quantity being determined but this may not necessarily be an accurate value"

Describe what is meant by

(i) accuracy

[1]

(ii) precision

[1]

Mark Scheme (2 marks):

(i) Accuracy is defined as a measure of:

- How close a measured / calculated value is to the true value; *[1 mark]*

(ii) Precision is defined as a measure of:

- How close the measured values are to one another **OR** the consistency of values obtained by repeated measurements; *[1 mark]*

[Total: 2 marks]

*Make sure not to confuse **precision** with **resolution** i.e. how many significant figures the measurement would be given to is to do with resolution, not precision*

Part (b) [3 marks]

Draw lines on Fig. 1.1 to show how accuracy and precision are affected by different types of error

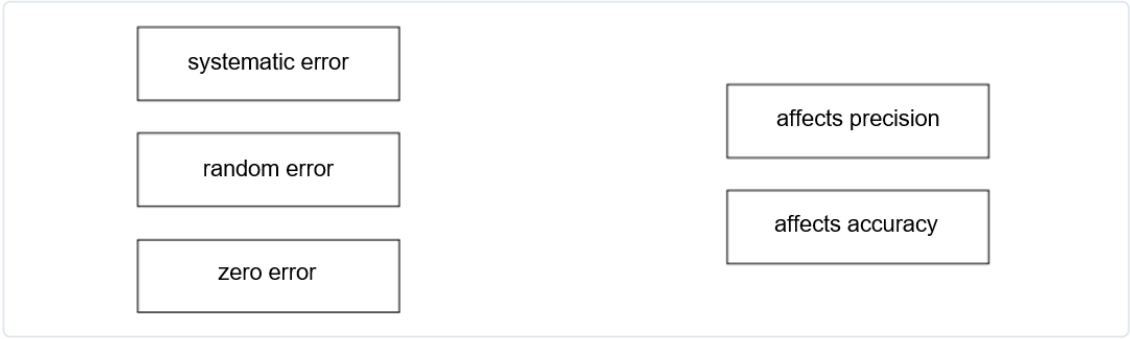
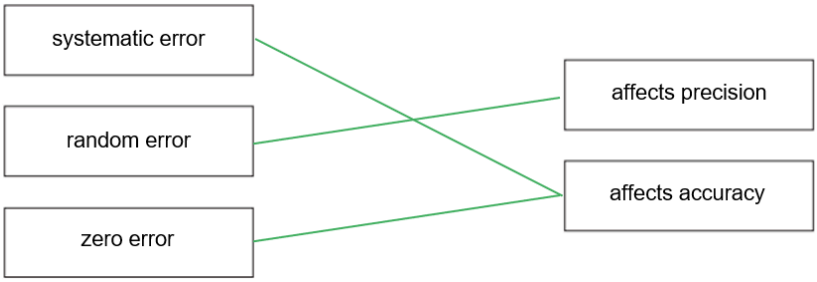


Fig. 1.1

Mark Scheme (3 marks):

The correct answers are:

One mark *for each correct line:*



[Total: 3 marks]

Part (c) **[4 marks]**

The teacher uses the diagram shown in Fig. 1.2 to illustrate the difference between accuracy and precision.

The circles represent targets A, B, C and D and the dots represent arrows hitting the targets.

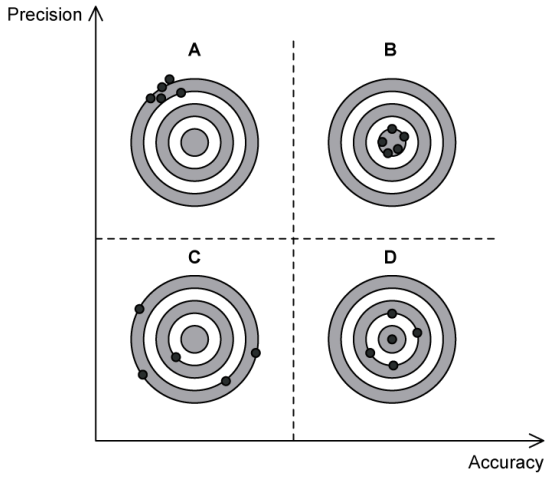


Fig. 1.2

Complete the following sentences to explain how targets A, B, C and D represent differing degrees of accuracy and precision.

neither accurate nor precise	accurate but not precise	precise but not accurate	both accurate and precise

A is .....

B is .....

C is .....

D is .....

Mark Scheme (4 marks):

The correct sentences are as follows:

- A is ***precise but not accurate***; *[1 mark]*
- B is ***both accurate and precise***; *[1 mark]*
- C is ***neither accurate nor precise***; *[1 mark]*
- D is ***accurate but not precise***; *[1 mark]*

[Total: 4 marks]

High precision *means a small spread of values*

- *This is shown by the dots located close together i.e. targets **A** and **B***

Low precision *means a large spread of values*

- *This is shown by the dots located far apart from one another i.e. **C** and **D***

High accuracy *means close to the true value*

- *This is shown by the dots located near the centre i.e. targets **B** and **D***

Low accuracy *means not close to the true value*

- *This is shown by the dots located away from the centre i.e. **A** and **C***

Part (d) [3 marks]

The temperature of the air in a room is measured using a mercury-in-glass thermometer.

For each of the following, tick [✓] one box to indicate whether the experimental technique would improve accuracy, precision or neither. The first row has been completed as an example.

	improves accuracy	**improves precision**	**improves neither**
correcting for an incorrectly calibrated thermometer	✓		
taking repeat readings of temperature and finding an average			
using a thermometer with a linear scale			
keeping your eye in line with the scale and the liquid level for each temperature reading			

Mark Scheme (3 marks):

The correct table is as follows:

	improves accuracy	improves precision	improves neither
correcting for an incorrectly calibrated thermometer	✓		
taking repeat readings of temperature and finding an average		✓	
using a thermometer with a linear scale			✓
keeping your eye in line with the scale and the liquid level for each temperature reading	✓		

- *One mark for each correct tick*

[Total: 3 marks]

*Remember: **precise** readings are ones that are close **together**:*

- *Repeat readings can give very similar measurements and can make identifying anomalous results much easier*
- *Reading the scale at eye level reduces parallax error and ensures that all readings are taken in the same way*

*Remember: **accurate** results are ones that are close to the **true value***

- *An incorrectly calibrated thermometer would make all the readings higher or lower by the same amount, hence it would make the final value higher or lower by a set amount*

Question 2 Easy [13 marks]

Part (a) [2 marks]

Distinguish between random and systematic errors.

Mark Scheme (2 marks):

Distinguish between random and systematic errors:

Systematic errors

*Any **one** from:*

- Cause all readings to differ from the true value by the same amount; *[1 mark]*
- Cannot be eliminated by averaging; *[1 mark]*
- Caused by errors / faults in measuring instrument / experimental set-up; *[1 mark]*

Random errors

*Any **one** from:*

- Cause readings to spread / scatter around the true value; *[1 mark]*
- Can be eliminated by averaging (only if not stated for systematic error); *[1 mark]*
- Caused by factors which vary in an unpredictable way **OR** caused by errors due to experimenter / observer; *[1 mark]*

[Total: 2 marks]

For random errors the most common wrong answer is 'human error'. This is a common phrase which we all understand, so it's tempting to say it here.

*However, in our physics experiments, it could be argued that **all** errors are due to human error. If the experimenter forgot to check their measuring scales were at zero before starting, the error introduced to the data is systematic, but the mistake was made by a human!*

*To avoid this, **never** say 'human error' in your physics exam. Be more specific about the nature of the problem.*

Part (b) [3 marks]

From a set of metal discs, one has a mass of 9.5 g and a thickness of approximately 2 mm.

State the instrument that should be used to measure

(i) the mass of one or more discs

[1]

(ii) the thickness of one disc

[1]

(iii) the thickness of a stack of 20 discs.

[1]

Mark Scheme (3 marks):

(i) The best instrument for measuring mass is:

- A digital / top-pan balance; *[1 mark]*

(ii) The best instrument for measuring the thickness of 1 disc is:

- Micrometer / screw gauge / digital callipers; *[1 mark]*

(iii) The best instrument for measuring the thickness of 20 discs is:

- Vernier callipers; *[1 mark]*

[Total: 3 marks]

It's useful to have an idea of the maximum possible readings of equipment as well as their lowest i.e. the resolution, so that you can make the best judgement for the most appropriate equipment to use in different scenarios

The line between when to use a micrometer or vernier callipers can get blurred, a good range to use for each is

- *Micrometer: 0.01 mm to 2.5 cm*
- *Vernier callipers: 0.1 mm to 15 cm*

For lengths between 10 - 30 cm, a 30 cm ruler is usually recommended, and anything above that should be measured using a metre rule

Part (c) **[5 marks]**

A student takes five repeat readings of the thickness of the disc. The readings are shown in Table 1.1.

Table 1.1

thickness / mm	2.012	2.001	1.998	2.008	1.994

Using the readings in Table 1.1, calculate

(i) the mean value of thickness

[2]

(ii) the range of the results

[1]

(iii) the uncertainty in the mean value of thickness.

[2]

Mark Scheme (5 marks):

(i) The mean value of thickness is:

- Mean = the sum of all the readings divided by the number of readings
- Mean = $\frac{2.012 + 2.001 + 1.998 + 2.008 + 1.994}{5}$ *[1 mark]*
- Mean = 2.0026 = 2.003 mm *[1 mark]*

(ii) The range of the results is:

- Range = the difference between the highest and lowest value
- Range = 2.012 − 1.994 = 0.018 mm *[1 mark]*

(iii) The uncertainty in the mean value of thickness is:

- Uncertainty in mean value = ± half the range
- Uncertainty in mean value = ± (½ × 0.018) *[1 mark]*
- Uncertainty in mean value = ± 0.009 mm *[1 mark]*

[Total: 5 marks]

Remember the mean value should always be quoted to the same number of ***significant figures*** *as the readings*

Part (d) **[3 marks]**

State how the instrument used to measure thickness can

(i) reduce or avoid a systematic error

[1]

(ii) reduce random errors

[2]

Mark Scheme (3 marks):

The instrument used to measure thickness is the micrometer:

(i) Reduce or avoid a systematic error by:

- Checking for a zero error / zeroing the micrometer before use; *[1 mark]*

(ii) Reduce random errors by:

- Taking (at least) three readings; *[1 mark]*
- At different points around the disc; *[1 mark]*

> ***[Total: 3 marks]***

Question 3 Easy [10 marks]

Part (a) [4 marks]

A student uses a stopwatch to measure the time taken for one complete swing of a pendulum.

They measure the time for 10 complete oscillations to be 16.4 ± 0.1 s.

Calculate

(i) the mean value of the time taken for one complete swing

[1]

(ii) the absolute uncertainty in this mean value

[1]

(iii) the percentage uncertainty in this mean value.

[2]

Mark Scheme (4 marks):

(i) Calculate the mean time for one complete swing:

- Time of 1 oscillation = $\frac{16.4}{10}$ = 1.64 s *[1 mark]*

(ii) Determine the absolute uncertainty:

- The absolute uncertainty must also be divided by 10
- Absolute uncertainty = $\pm$ $\frac{0.1}{10}$ = ± 0.01 s *[1 mark]*

(iii) Calculate the percentage uncertainty:

- % uncertainty = $\frac{uncertainty}{value} \times 100\%$
- % uncertainty = $\frac{0.01}{1.64} \times 100\%$ *[1 mark]*
- % uncertainty = 0.6098 = 0.61% *[1 mark]*

  ∴ mean time period *T* = 1.64 ± 0.01 s or 1.64 s ± 0.61% (2 s.f.)

[Total: 4 marks]

The number of significant figures you quote calculated uncertainties to is very important

*A significant figure is a reliably **known** digit, whereas an uncertainty is an **estimate**. Therefore, the percentage uncertainty should be given to the **same** number of significant figures as the result*

*And remember, uncertainties should **never** be given to **more** significant figures than the result as it **cannot** be known to a greater degree of precision than the result itself*

Part (b) [2 marks]

The student is using the pendulum to determine a value for the acceleration of free fall *g*.

As well as the period *T* of oscillation, they also take measurements of the length *L* of the pendulum.

The value obtained, with its uncertainty, is *L* = 67 ± 1 cm.

Calculate the percentage uncertainty in the measurement of the length *L*.

Mark Scheme (2 marks):

Calculate the percentage uncertainty in the measurement of *L*:

- % uncertainty = $\frac{uncertainty}{value} \times 100\%$
- % uncertainty = $\frac{1}{67} \times 100\%$ *[1 mark]*
- % uncertainty = 1.4925 = 1.5% *[1 mark]*

  ∴ length *L* = 67 ± 1 cm or 67 s ± 1.5% (2 s.f.)

[Total: 2 marks]

Part (c) [2 marks]

The relationship between *T*, *L* and *g* is given by

$$g = 4\pi^2 \frac{L}{T^2}$$

Using your answers in **(a)** and **(b)**, show that the percentage uncertainty in the value of *g* is 2.7%.

Mark Scheme (2 marks):

Show that the percentage uncertainty in the value of *g* is 2.7%:

List the known quantities:

- % uncertainty in *L* = 1.5%
- % uncertainty in *T* = 0.6%

Calculate the % uncertainty in *g*:

- % uncertainty in *g* = % uncertainty in *L* + (2 × % uncertainty in *T*) *[1 mark]*
- % uncertainty in *g* = 1.5 + (2 × 0.6) *[1 mark]*
- % uncertainty in *g* = 2.7 %

[Total: 2 marks]

Make sure you learn the rules for calculating uncertainties

In the relationship $g = 4\pi^2 \frac{L}{T^2}$ *there is division and T is raised to a power, and the factor* $4\pi^2$ *can be ignored (since it is a ratio not a measurement), so the rules for calculating the uncertainty in g will involve:*

- *Adding the % uncertainties together - this applies when multiplying or dividing data*
- *Multiply the % uncertainty by the power - this only applies to quantities that are raised to a power*

Part (d) [2 marks]

There are some occasions where the resolution of an instrument is not the only limiting factor of uncertainty in a measurement.

State

(i) the meaning of the 'resolution of an instrument'

[1]

(ii) **one** limiting factor that affects the uncertainty in the measurement of time using a stopwatch.

[1]

Mark Scheme (2 marks):

(i) Resolution of an instrument is defined as:

- The smallest observable / recognisable change in a reading (on the instrument); *[1 mark]*

(ii) The limiting factor of a stopwatch is:

- The reaction time of the observer / experimenter; *[1 mark]*

[Total: 2 marks]

Question 4 Medium [4 marks]

Part (a) [2 marks]

The speed *v* of a transverse wave on a uniform string is given by the expression

$$v = \sqrt{\frac{Tl}{m}}$$

where *T* is the tension in the string, *l* is its length and *m* is its mass.

An experiment is performed to determine the speed *v* of the wave. The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement	uncertainty
T	1.8 N	± 5%
l	126 cm	± 1%
m	5.1 g	± 2%

Use the data in Table 1.1 to calculate the speed *v*.

v = ... m s^{-1}

Mark Scheme (2 marks):

Use the data in Table 1.1 to calculate the speed v:

List the known quantities and convert into SI base units:

- T = 1.8 N
- l = 126 cm = 126 × 10^{-2} m
- m = 5.1 g = 5.1 × 10^{-3} kg

Substitute the known quantities into the equation given:

- $v = \sqrt{\frac{Tl}{m}}$
- $v = \sqrt{\frac{1.8 \times 126 \times 10^{-2}}{5.1 \times 10^{-3}}}$ *[1 mark]*
- v = 21.1 m s^{-1} = 21 m s^{-1} (2 s.f.) *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

Use your answer in **(a)** and the data in Table 1.1 to calculate the absolute uncertainty in the value of *v*.

absolute uncertainty = ... m s^{-1}

Mark Scheme (2 marks):

Calculate the absolute uncertainty in the value of v:

List the known quantities:

- Uncertainty in T = ± 5%
- Uncertainty in l = ± 1%
- Uncertainty in m = ± 2%

Calculate the percentage uncertainty in v:

- From the equation $v = \sqrt{\frac{Tl}{m}} = \frac{Tl}{m}^{\frac{1}{2}}$
- % uncertainty in v = ½ × (5% + 1% + 2%) = 4 % *[1 mark]*

Calculate the absolute uncertainty in v:

- Absolute uncertainty in $v = 4\% \times v = 0.04 \times 21$
- Absolute uncertainty in $v = 0.84$ m s^{-1} *[1 mark]*

[Total: 2 marks]

- *T and l are multiplied together and divided by m*
  - *So, their percentage uncertainties are added together*
  - *5% + 1% + 2% = 8%*
- *The result is then put to the power of ½*
  - *So the percentage uncertainty is halved*
  - *8% × ½ = 4%*
- *Learn the **rules** for **combining uncertainties** from the **revision notes***

Question 5 Medium [18 marks]

Part (a) [4 marks]

A student used the apparatus shown to investigate the time taken for a hollow cylinder to roll down a ramp as shown in Fig. 1.1.

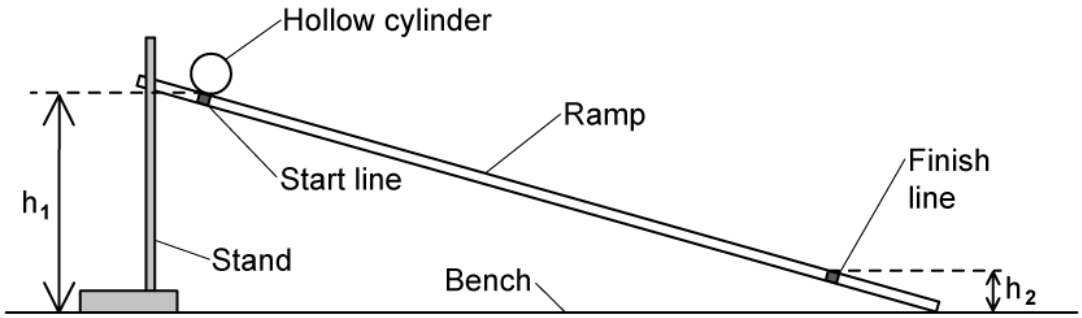


Fig. 1.1

State the most appropriate instrument for the measurement of the following, and describe how they can ensure the measurements are as accurate as possible

(i) the heights h_1 and h_2

[2]

(ii) the time taken for the cylinder to roll down the ramp

[2]

Mark Scheme (4 marks):

(i) To measure heights h_1 and h_2:

- Use a metre ruler; *[1 mark]*

To ensure the measurements are as accurate as possible are:

*Any **one** from:*

- Place the rule as close as possible to the ramp; *[1 mark]*
- Use a set square / spirit level to ensure the rule is vertical; *[1 mark]*
- Ensure the rule reads zero at the bench; *[1 mark]*
- Avoid parallax error e.g. use a set square / read the scale at eye level; *[1 mark]*

(ii) To measure the time taken:

- Use a timer / stopwatch; *[1 mark]*

To ensure the measurements are as accurate as possible are:

*Any **one** from:*

- Start the timer immediately once the cylinder is released; *[1 mark]*
- Use a (fiducial) marker to mark the endpoint (to stop the timer at the same time); *[1 mark]*
- Repeat the experiment several times and take an average; *[1 mark]*

[Total: 4 marks]

Part (b) **[2 marks]**

The student wants to find the difference in height Δ*h* between the start line and finish line

The heights are recorded as

$$h_1 = 67.5 \pm 0.1 \text{ cm}$$

$$h_2 = 8.6 \pm 0.1 \text{ cm}$$

Determine the change in height Δ*h*, quoting your answer with its uncertainty

Δ*h* = ... ± ..... cm

Mark Scheme (2 marks):

The change in height Δ*h* is:

- $\Delta h = h_1 - h_2 = 67.5 - 8.6 = 58.9$ cm

The uncertainty in Δ*h* is:

- The uncertainty in each measurement is 0.1 cm
- Since the values of *h* are subtracted, the uncertainty is 0.1 cm + 0.1 cm = 0.2 cm *[1 mark]*

Quote Δ*h* with its uncertainty:

- Δ*h* = 58.9 ± 0.2 cm *[1 mark]*

[Total: 2 marks]

Part (c) **[6 marks]**

The student placed the cylinder on the start line and released it. She measured the time *t* for the cylinder to roll to the finish line. She repeated the measurements several times as shown in Table 1.1.

Table 1.1

***t* / s**	0.88	0.82	0.80	0.86

(i) Use the data in Table 1.1 to calculate the mean value of t and its uncertainty.

mean value of t = ... ± ..... [2]

(ii) Comment on the accuracy and precision of the data the student collected.

[2]

(iii) Explain how reducing the value of h_1 could improve the measurement of t.

[2]

Mark Scheme (6 marks):

(i) Calculate the mean value of t and its uncertainty

- Mean value of t: $\frac{0.88 + 0.82 + 0.80 + 0.86}{4} = 0.84$ s *[1 mark]*
- Uncertainty in t: $\pm\frac{1}{2}$ $range$ $= \pm\frac{1}{2}$ 0.88 - 0.80 = ± 0.04 s *[1 mark]*

(ii) Comment on the accuracy and precision of the data:

- Precision describes how close a set of measurements are to one another
- The readings are precise as they are reasonably close together; *[1 mark]*
- Accuracy describes how close the readings are to the true value
- The level of accuracy of the data cannot be determined (yet) **OR** it cannot be told whether the readings are close to the true value (yet); *[1 mark]*

(iii) Reducing h_1 might improve the measurement of t because:

- The values of t will increase **OR** the cylinder will move more slowly; *[1 mark]*

*As well as any **one** from:*

- So the percentage uncertainty in t will reduce; *[1 mark]*
- It will be easier to judge when the cylinder crosses the finish line; *[1 mark]*
- The effect of reaction time will be reduced; *[1 mark]*

[Total: 6 marks]

Part (d) **[6 marks]**

The relationship between t and the acceleration of free fall g is given by

$$t^2 = \frac{4s^2}{g\Delta h}$$

where s is the distance along the ramp between the start line and the finish line.

The student recorded the ramp length as s = 102.0 cm ± 0.1 cm

(i) Determine a value for g.

[2]

(ii) Determine the percentage uncertainty in the value of g.

[2]

(iii) Deduce whether the value of g is accurate.

[2]

Mark Scheme (6 marks):

List the known quantities:

- Change in height, Δh = 58.9 ± 0.2 cm
- Time taken, t = 0.84 ± 0.04 s
- Ramp length, s = 102.0 ± 0.1 cm

(i) Determine a value for g:

- $t^2 = \frac{4s^2}{g\Delta h} \quad \Rightarrow \quad g = \frac{4s^2}{t^2 \Delta h}$
- $g = \frac{4 \times 102 \times 10^{-2}{}^2}{0.84^2 \times 58.9 \times 10^{-2}}$ *[1 mark]*
- g = 10.014 = 10.0 m s^{-2} *[1 mark]*

(ii) Determine the percentage uncertainty in the value of g

- When multiplying data, add the fractional or percentage uncertainties
- When a quantity is raised to a power, multiply the uncertainty by the power
- Fractional uncertainty in g: $\frac{\Delta g}{g} = 2 \times \frac{\Delta s}{s} + \frac{\Delta \Delta h}{\Delta h} + 2 \times \frac{\Delta t}{t}$
- % uncertainty in $g = 2\frac{0.1}{102} + \frac{0.2}{58.9} + 2\frac{0.04}{0.84} \times 100\%$ *[1 mark]*
- % uncertainty in g = 10.059 = 10 % *[1 mark]*

(iii) Deduce whether the measured value of g is accurate

Calculate the lower limit of g:

- Lower limit of g: $g \times (1 - 0.10) = 10 \times 0.90 = 9.0$ m s^{-2} *[1 mark]*

Write a conclusion:

- The measured value of g is 10.0 ± 1.0 m s^{-2} and the accepted value of g is 9.81 m s^{-2} hence, the value is accurate (since this lies within the lower limit of g); *[1 mark]*

> ***[Total: 6 marks]***
>
> *When you see command words such as 'deduce', make sure to write a **conclusion** at the end in order to score top marks*
>
> *Knowing how to carry out the calculation is important, but examiners also want to see if you understand what you have calculated!*

Question 6 Medium [15 marks]

Part (a) [3 marks]

A cube made from aluminium is placed on a horizontal surface, as shown in Fig. 1.1.

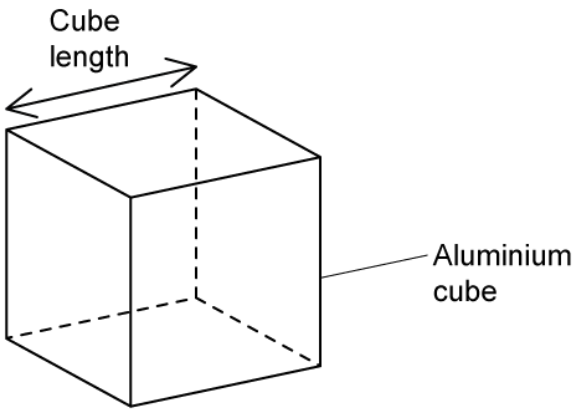


Fig. 1.1

The following measurements were made on the cube:

mass = 0.59 ± 0.01 kg

length of one side = 6.0 ± 0.1 cm

Calculate the pressure produced by the cube on the surface.

Mark Scheme (3 marks):

Calculate the pressure produced by the cube on the surface:

List the known quantities:

- Mass, m = 0.59 kg
- Length of cube, L = 6.0 cm = 0.06 m

Determine the pressure from the force and the area in contact with surface:

- Force on surface, F = weight = mg = 0.59 × 9.81

  AND

- Area in contact with surface, $A = L^2 = (0.06)^2$ *[1 mark]*
- Pressure: $P = \frac{F}{A}$
- $P = \frac{mg}{L^2} = \frac{0.59 \times 9.81}{0.06^2}$ *[1 mark]*
- P = 1607.75 = 1610 Pa (3 s.f.) *[1 mark]*

[Total: 3 marks]

Part (b) **[4 marks]**

(i) Calculate the percentage uncertainty in the pressure.

[2]

(ii) State the pressure, with its actual uncertainty.

pressure = ... ± ..... Pa [2]

Mark Scheme (4 marks):

List the known quantities:

- Mass, m = 0.59 ± 0.01 kg
- Cube length, L = 6.0 ± 0.1 cm
- Pressure, P = 1600 Pa (from **(a)**)

(i) Calculate the percentage uncertainty in the pressure:

- When multiplying data, add the fractional or percentage uncertainties
- When a quantity is raised to a power, multiply the uncertainty by the power
- Pressure: $P = \frac{mg}{L^2}$
- Fractional uncertainty in P: $\frac{\Delta P}{P} = \frac{\Delta m}{m} + 2 \times \frac{\Delta L}{L}$
- % uncertainty in P = $\frac{0.01}{0.59} + 2\frac{0.1}{6.0} \times 100\%$ *[1 mark]*
- % uncertainty in P = 5.028 = 5 % *[1 mark]*

(ii) State the pressure, with its actual uncertainty:

- Actual uncertainty in P = 1600 × 0.05 = 80 Pa *[1 mark]*
- Pressure = 1600 ± 80 Pa *[1 mark]*

Other acceptable answers for pressure = 1610 ± 80 or 1608 ± 81 Pa

[Total: 4 marks]

Part (c) **[4 marks]**

Calculate the density of aluminium with its uncertainty.

Express your answer to an appropriate number of significant figures.

density ..... ± ..... g cm^{-3}

Mark Scheme (4 marks):

Calculate the density of aluminium, in g cm^{-3}, with its uncertainty:

List the known quantities:

- Mass, m = 0.59 ± 0.01 kg = 590 ± 10 g
- Cube length, L = 6.0 ± 0.1 cm

Calculate the density in g cm^{-3}:

- Density: $\rho = \frac{m}{V}$
- $\rho = \frac{m}{L^3} = \frac{590}{6.0^3}$
- ρ = 2.731 = 2.7 g cm^{-3} *[1 mark]*

Determine the uncertainty in density:

- Uncertainty in ρ: $\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 3 \times \frac{\Delta L}{L}$
- $\frac{\Delta\rho}{\rho} = \frac{10}{590} + 3\frac{0.1}{6.0}$ *[1 mark]*
- $\frac{\Delta\rho}{\rho}$ = 0.0669 = 0.07 *[1 mark]*

Quote the density and uncertainty to an appropriate amount of sig figs:

- Actual uncertainty in ρ: $\Delta\rho$ = 2.73 × 0.07 = 0.18 = 0.2 g cm^{-3}
- Density = 2.7 ± 0.2 g cm^{-3} *[1 mark]*

[Total: 4 marks]

Remember, the uncertainty must always be quoted to at least 1 sig fig less than the quantity itself - this is because you cannot know the uncertainty in a quantity to a greater precision that the quantity itself!

Part (d) **[4 marks]**

The student uses vernier callipers to measure the length of the side of the cube. After completing the experiment, he realises that when the callipers are fully closed, the reading is not zero.

(i) State and explain whether this introduces a random error or a systematic error in the readings of the length.

[2]

(ii) Explain why the readings are precise but not accurate.

[2]

Mark Scheme (4 marks):

(i) The type of error is:

- Systematic; *[1 mark]*

- (Because) all readings have the same error **OR** this error cannot be eliminated by repeating and averaging; *[1 mark]*

(ii) The readings are precise but not accurate because:

- Vernier callipers measure to a fraction of a millimeter / resolution of 0.1 mm, so the readings are precise **OR** if repeated, readings will be consistent / almost constant; *[1 mark]*
- But, all readings will have the same error, hence they are not accurate; *[1 mark]*

[Total: 4 marks]

Question 7 Medium [18 marks]

Part (a) [4 marks]

Analogue voltmeters and cathode-ray oscilloscopes (c.r.o.) are both devices which can measure potential differences.

For the analogue voltmeter and the c.r.o., describe **one** example of:

(i) a systematic error that can affect both devices

[1]

(ii) a systematic error that can only affect the analogue voltmeter

[1]

(iii) a random error that can affect both devices

[1]

(iv) a random error that can only affect the analogue voltmeter.

[1]

Mark Scheme (4 marks):

(i) A systematic error that can affect both devices is:

- Incorrect calibration (of the device); *[1 mark]*

(ii) A systematic error that can only affect the analogue voltmeter is:

- Zero error; *[1 mark]*

(iii) A random error that can affect both devices is:

- Incorrect interpolation between scale readings / divisions; *[1 mark]*

(iv) A random error that can only affect the analogue voltmeter is:

- Parallax error / incorrectly reading the scale from different angles; *[1 mark]*

[Total: 4 marks]

Part (b) [6 marks]

The potential difference across a resistor is measured using the analogue voltmeter shown in Fig. 1.1.

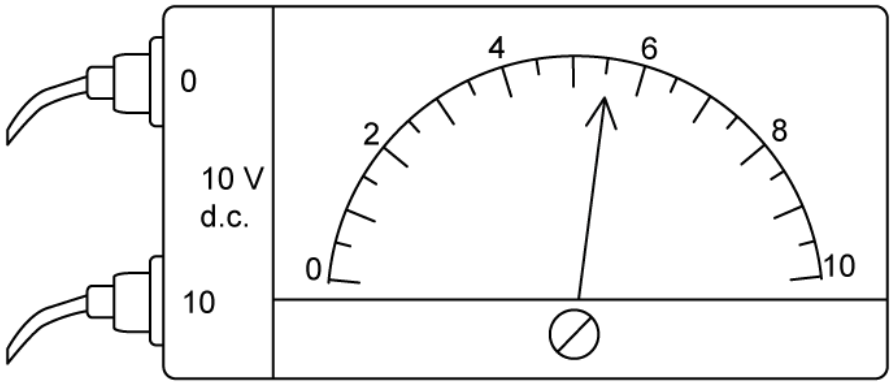


Fig. 1.1

The resistor is labelled as having a resistance of 125 Ω ± 3%.

(i) Calculate the power dissipated by the resistor.

[2]

(ii) Calculate the percentage uncertainty in the calculated power.

[2]

(iii) Determine the value of the power, with its absolute uncertainty, to an appropriate number of significant figures.

power = ... ± ..... W [2]

Mark Scheme (6 marks):

(i) Calculate the power dissipated by the resistor:

- (From Fig. 1.1): voltmeter reading = 5.5 V *[1 mark]*
- Electrical power: $P = \frac{V^2}{R}$
- $P = \frac{5.5^2}{125}$ = 0.242 W *[1 mark]*

(ii) Calculate the percentage uncertainty in the calculated power:

- Uncertainty in voltmeter reading = half the smallest division = ± $\frac{1}{2}$ × 0.5 = ± 0.25 V
- % uncertainty in voltmeter reading = $\frac{0.25}{5.5} \times 100\%$ = 4.5% *[1 mark]*
- % uncertainty in power = (2 × 4.5%) + 3% = 12 % *[1 mark]*

(iii) The power with its uncertainty is:

- Absolute uncertainty in P = 0.12 × 0.242 = ± 0.029 W *[1 mark]*

- Power = 0.24 ± 0.03 W (2 s.f.) *[1 mark]*

[Total: 6 marks]

Part (c) **[5 marks]**

A microphone connected to a cathode ray oscilloscope (c.r.o.) detects a sound wave of frequency *f*. The signal from the microphone is observed on the c.r.o. as shown in Fig. 1.2.

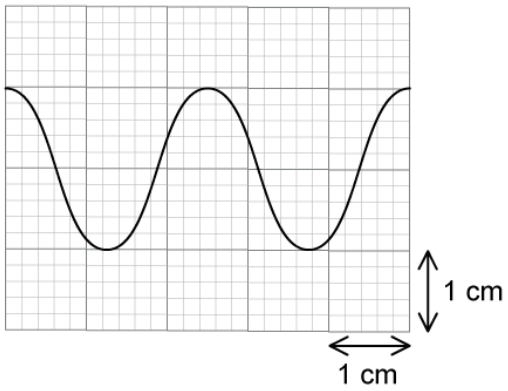


Fig. 1.2

The time-base setting of the c.r.o. is 0.50 ms cm^{-1}. The Y-plate setting is 2.5 mV cm^{-1}.

Using the trace in Fig. 1.2, determine

(i) the amplitude of the signal, in mV

amplitude = ... mV [2]

(ii) the frequency, in Hz

frequency = ... Hz [3]

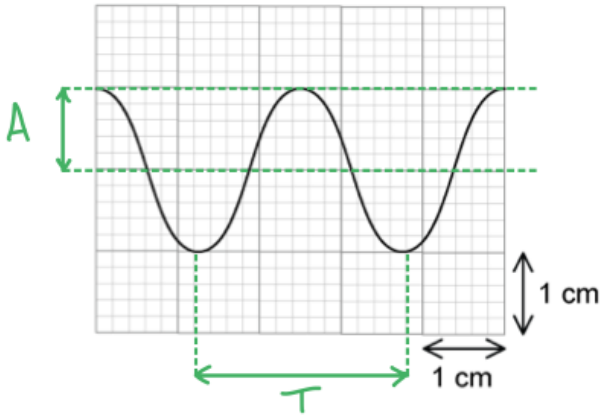
Mark Scheme (5 marks):

(i) Use the trace to determine the amplitude:

- Y-plate setting = 2.5 mV cm^{-1}
- Amplitude is measured from the mid-point to the peak of the wave
- (From trace): amplitude reading = 1 large square = 1 cm *[1 mark]*
- Amplitude, A = 1 × 2.5 = 2.5 mV *[1 mark]*

(ii) Use the trace to determine the frequency:

- Time-base setting = 0.50 ms cm^{-1}
- Time period is measured from one peak to the next peak
- (From trace): time period reading = 13 small squares = 13 × 0.2 = 2.6 cm *[1 mark]*
- Time period, T = 2.6 × (0.5 × 10^{-3}) = 0.0013 s *[1 mark]*
- Therefore, frequency, $f = \frac{1}{T} = \frac{1}{0.0013}$ = 769 = 770 Hz *[1 mark]*



[Total: 5 marks]

Part (d) [3 marks]

(i) Determine the percentage uncertainty in amplitude caused by reading the scale on the c.r.o.

[2]

(ii) State the amplitude with its actual uncertainty.

amplitude = ... ± ..... mV [1]

Mark Scheme (3 marks):

(i) Determine the % uncertainty in amplitude:

- Y-plate setting = 2.5 mV cm^{-1}
- 1 division = 0.2 cm
- Uncertainty in a reading = half the smallest division = ± $\frac{1}{2}$× (0.2 × 2.5) = ± 0.25 mV *[1 mark]*
- % uncertainty in amplitude = $\frac{0.25}{2.5} \times 100\%$ = 10% *[1 mark]*

(ii) The amplitude, with its uncertainty, is:

- Amplitude = 2.50 ± 0.25 mV *[1 mark]*

[Total: 3 marks]

Question 8 Medium [10 marks]

Part (a) [3 marks]

A student participates in an experiment to measure g, the acceleration due to gravity on Earth. This is done using a simple pendulum.

The student suggests the period of oscillation T is related to the length L of the pendulum by the equation:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Table 1.1 shows the ten repeat readings of period T.

Table 1.1

0.67	0.66	0.67	0.68	0.69	0.64	0.66	0.65	0.68	0.65

Determine the mean period of oscillation and its percentage uncertainty.

Mark Scheme (3 marks):

Determine the mean period of oscillation and its percentage uncertainty:

- Mean $T = \frac{0.67+0.66+0.67+0.68+0.69+0.64+0.66+0.65+0.68+0.65}{10}$
- Mean T = 0.67 s *[1 mark]*

Calculate the absolute uncertainty in *T*:

- Uncertainty in T = ± half the range
- Uncertainty in T = ± $\frac{1}{2}$ (0.69 – 0.64) = 0.025 s
- Uncertainty in T = ±0.03 s *[1 mark]*

Calculate the percentage uncertainty in *T*:

- Percentage uncertainty in $T = \frac{0.03}{0.67} \times 100$
- Percentage uncertainty in T = 4.478 = ±4% *[1 mark]*
- Therefore, T = 0.67 s ± 4%

[Total: 3 marks]

Part (b) [3 marks]

In a new experiment, the length L of the pendulum is measured with a percentage uncertainty of 1.8% and the acceleration due to free-fall g is measured with a percentage uncertainty of 1.6%.

If the time for the pendulum to complete 20 oscillations is 18.4 s, determine the time period for one oscillation and the absolute uncertainty in this value.

Mark Scheme (3 marks):

List the known quantities:

- Uncertainty in L, ΔL = 1.8%
- Uncertainty in g, Δg = 1.6%

Calculate the time period, T:

- Time period, T = $\frac{\text{total time of oscillations}}{\text{number of oscillations}}$
- $T = \frac{18.4}{20}$ = 0.92 s *[1 mark]*

Calculate the absolute uncertainty in T:

- $T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\frac{L^{\frac{1}{2}}}{g}$
- % uncertainty in $T = \frac{1}{2} \times 1.8 + \frac{1}{2} \times 1.6$ = 1.7% *[1 mark]*
- Uncertainty in T = 0.017 × 0.92 = 0.02 s (1 s.f) *[1 mark]*
- Therefore, T = 0.92 ± 0.02 s

[Total: 3 marks]

Break this question down into stages and make sure you propagate the uncertainty correctly.

When propagating the uncertainty containing a square root of a quantity, remember that the square root of a number is the same as putting it to the power of ½.

Part (c) **[2 marks]**

Measurements of time periods for different lengths of pendula were taken using a stopwatch and plotted on the graph shown in Fig. 1.1.

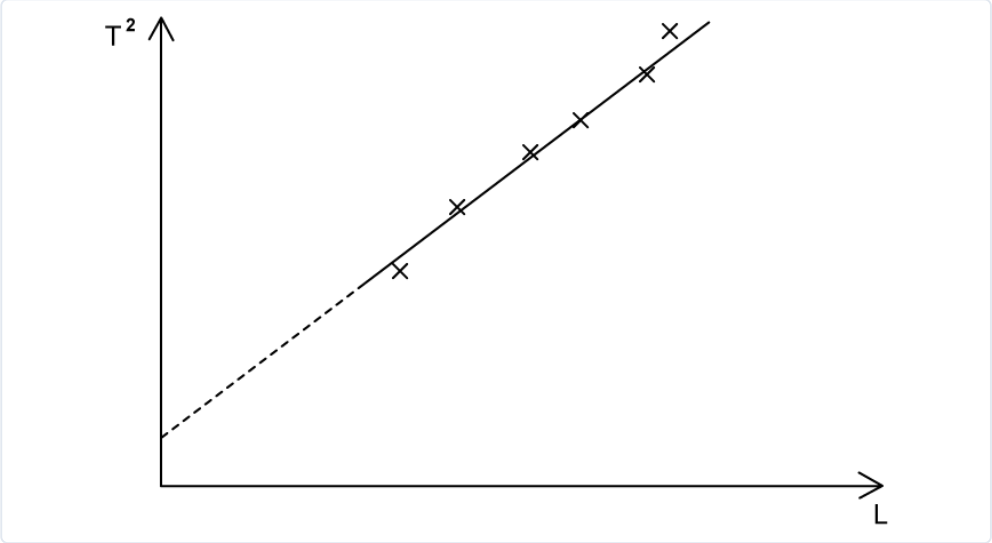


Fig. 1.1

Explain how the graph in Fig. 1.1 indicates that the readings are subject to systematic and random uncertainties.

Mark Scheme (2 marks):

The readings are subject to systematic uncertainties because:

- The graph has a y-intercept; *[1 mark]*

The readings are subject to random uncertainties because:

- Points are scattered around the line of best fit; *[1 mark]*

[Total: 2 marks]

Part (d) **[2 marks]**

The period T for a mass m hanging on a spring performing simple harmonic motion is given by the equation:

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Such a system is used to determine the spring constant k. The fractional error in the measurement of the period T is α and the fractional error in the measurement of the mass m is β.

Determine the fractional error in the calculated value of k in terms of α and β.

Mark Scheme (2 marks):

Determine the fractional error in k:

List the known quantities:

- Fractional error in T, $\frac{\Delta T}{T} = \alpha$
- Fractional error in m, $\frac{\Delta m}{m} = \beta$

Write an expression for the uncertainty of k in terms of α and β:

- Period of a mass-spring: $T = 2\pi\sqrt{\frac{m}{k}} \quad \Rightarrow \quad k = 4\pi^2\frac{m}{T^2}$ *[1 mark]*
- $\frac{\Delta k}{k} = 2\frac{\Delta T}{T} + \frac{\Delta m}{m}$
- $\frac{\Delta k}{k} = 2\alpha + \beta$ *[1 mark]*

[Total: 2 marks]

Remember the rules for propagating errors. These must be applied carefully to the rearranged form of this equation. These are:

- *Values that are added or subtracted:*
  - *Add their **absolute** uncertainties*
  - *In this example, there are no quantities that are added or subtracted*
- *Values that are multiplied or divided:*

- *Add their **percentage** or **fractional** uncertainties*
- *In this example, m and T are divided so we can add %ΔT + %Δm, or the fractional uncertainties (as shown in the mark scheme)*

- *Values that are to a power:*
  - *Multiply their **percentage** or **fractional** uncertainty by the power*
  - *In this example, T is squared, so its uncertainty is 2ΔT*

Question 9 Medium [6 marks]

Part (a) [2 marks]

The drag force F acting on a car moving with speed v along a straight horizontal road is given by

$$F = k\rho A v^n$$

where n is an integer, ρ is the density of air, A is the cross-sectional area of the car, and k is a constant with no units.

An experiment is performed to determine the drag force F on the car. The measurements are shown in Table 1.1.

Table 1.1

quantity	measurement	uncertainty
ρ	1.20 kg m^{-3}	± 2%
A	5.1 m^2	± 4%
v	13.4 m s^{-1}	± 1%

Use SI base units to show that n is equal to 2.

Mark Scheme (2 marks):

To show that n is equal to 2:

- List the known SI base units:
  - Mass m = kg
  - Acceleration a = m s^{-2}
  - Density ρ = kg m^{-3}
  - Area A = m^2
  - Velocity v = m s^{-1}
- Determine the units of F:

$$F = ma$$

F = kg m s^{-2} *[1 mark]*

- Check the homogeneity of the units on both sides of the equation:

$$F = k\rho A v^n$$

$$\text{kg m s}^{-2} = \text{kg m}^{-3} \times \text{m}^2 \times \text{m s}^{-1}{}^n$$

$$\text{m}^n \text{ s}^{-n} = \frac{\text{kg m s}^{-2}}{\text{kg m}^{-3} \times \text{m}^2}$$

$$\text{m}^n \text{ s}^{-n} = \frac{\text{kg m s}^{-2}}{\text{kg m}^{-1}}$$

$$\text{m}^n \text{ s}^{-n} = \text{m}^2 \text{ s}^{-2}$$

$$n = 2 \textit{ [1 mark]}$$

Part (b) **[4 marks]**

The value of k for the car in **(c)** is 0.28. The uncertainty in k is negligible.

(i) Use the data in Table 1.1 to calculate the percentage uncertainty in F.

[1]

(ii) Determine F with its absolute uncertainty. Give your values to the appropriate number of significant figures.

[3]

Mark Scheme (4 marks):

(i) To calculate the percentage uncertainty in F:

- List the known quantities
  - Uncertainty in ρ = ± 2%
  - Uncertainty in A = ± 4%
  - Uncertainty in v = ± 1%
- Add together the percentage uncertainties:

% uncertainty in F = 2% + 4% + (2 × 1%)

% uncertainty in F = ± *8% [1 mark]*

From the equation $F = k\rho A v^2$, the quantities ρ, A, and v are multiplied together, so their percentage uncertainties are added together. k can be ignored as its uncertainty is negligible. Since v is squared, its percentage uncertainty must be multiplied by 2.

(ii) To determine F with its absolute uncertainty:

- List the known quantities
  - Density of air, ρ = 1.20 kg m^{-3}
  - Cross-sectional area of car, A = 5.1 m^2

- Speed of car, v = 13.4 m s^{-1}
- Constant, k = 0.28

- Calculate the drag force F:

$$F = k\rho A v^2$$

$$F = 0.28 \times 1.20 \times 4.9 \times 13.4^2$$

$$F = 307.69...$$

$$F = 308 \text{ N } \textit{[1 mark]}$$

- Calculate the absolute uncertainty in F:

$$\text{uncertainty in } F = 0.08 \times 308$$

$$\text{uncertainty in } F = 24.64$$

$$\text{uncertainty in } F = \pm 24.6 \text{ N } \textit{[1 mark]}$$

- State F with its absolute uncertainty to the appropriate number of significant figures:

$$F = 310 \pm 20 \text{ N } \textit{[1 mark]}$$

The appropriate number of significant digits of a final value is related to precision. The percentage uncertainty of 8% is what decides the precision in the final value of F. The rule is to round the absolute uncertainty to one significant figure, (20) and then match the precision of the final value to this (the tens place value, 310). This ensures we don't claim more precision in our final result than we actually have.

Question 10 Hard [8 marks]

Part (a) [2 marks]

In an experiment to determine the emissivity of a circular surface, the diameter *d* of the object is measured using the vernier scale shown in Fig. 1.1.

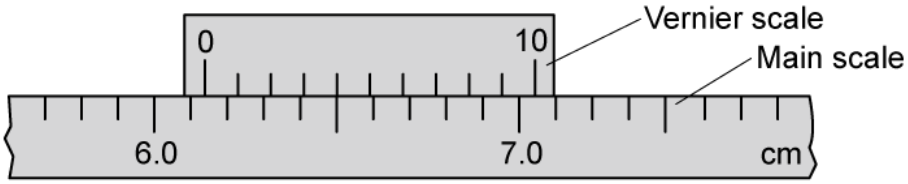


Fig. 1.1

Determine the diameter of the circular object from the reading in Fig. 1.1 and its associated uncertainty.

Mark Scheme (2 marks):

Determine the diameter of the circular object and its associated uncertainty:

- Main scale reading = 6.1 cm
- Vernier scale reading = 0.04 cm

Calculate the final reading by adding the two readings:

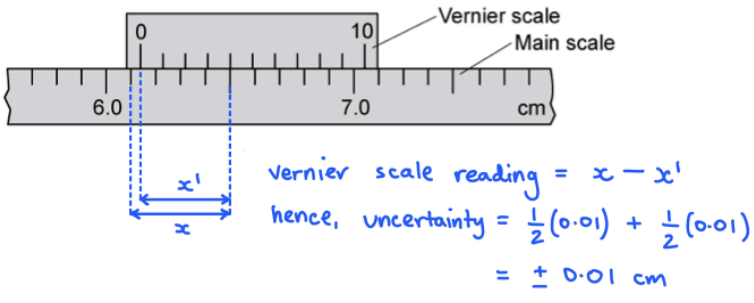
- Diameter, *d* = 6.1 + 0.04 = 6.14 cm *[1 mark]*

Determine the uncertainty in the diameter:

- Uncertainty in the measurement = ± smallest division
- The smallest division on the scale is 0.01 cm
- Hence, uncertainty in *d* = ± 0.01 cm *[1 mark]*

[Total: 2 marks]

Measurements on instruments such as a rulers or vernier scales involve a subtraction between two points on the scale. The vernier part of the reading is really a subtraction, so we use ± (smallest division) rather than ± (½ × smallest division)



That means 0.01 cm is the smallest scale reading that can be made without estimation, hence the reading above is 6.14 ± 0.01 cm

Part (b) [3 marks]

The energy per unit time *P* radiated by an object with surface area *A* at absolute temperature *T* is given by

$$P = e\sigma AT^4$$

where *e* is the emissivity of the object and *σ* is the Stefan-Boltzmann constant.

Show that the emissivity *e* has no unit.

Mark Scheme (3 marks):

Show that the emissivity *e* has no unit:

- Units on RHS:
  - $P = W$ *[1 mark]*
- Units on LHS:

  All correct for ***one mark***
  - Units of $T^4 = K^4$
  - Units of $A = m^2$
  - Units of $\sigma = W\ m^{-2}\ K^{-4}$

Compare the units on both sides:

- $P = e\sigma AT^4$
- $W = W\ m^{-2}\ K^{-4}\ m^2\ K^4$ *[1 mark]*
- RHS = LHS, hence *e* is unitless

[Total: 3 marks]

Part (c) [3 marks]

As well as diameter *d*, the following measurements were taken:

P = 3.0 ± 0.1 W

T = 500 ± 1 K

Determine the value of the emissivity *e* of the surface and its uncertainty. Give your answer to an appropriate degree of precision.

Mark Scheme (3 marks):

Determine the value of the emissivity *e* of the surface and its uncertainty:

List the known quantities:

- Power, P = 3.0 ± 0.1 W
- Temperature, T = 500 ± 1 K
- Diameter, d = 6.14 ± 0.01 cm = 0.0614 ± 0.0001 m (from **(a)**)
- Stefan-Boltzmann constant, $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

Determine the value of emissivity, *e*:

- Cross-sectional area of a circle: $A = \frac{\pi d^2}{4}$
- $P = e\sigma AT^4 = e\sigma\frac{\pi d^2}{4}T^4$
- $e = \frac{4P}{\pi\sigma d^2 T^4}$
- $e = \frac{4 \times 3.0}{\pi \times (5.67 \times 10^{-8}) \times (0.0614^2) \times (500^4)}$
- e = 0.2859 = 0.29 (2 s.f.) *[1 mark]*

Determine the uncertainty in emissivity, Δ*e*:

- Separate the constants and the quantities:
  - $e = \frac{4}{\pi\sigma}\frac{P}{d^2 T^4}$
- Propagate as fractional uncertainties:
  - $\frac{\Delta e}{e} = \frac{\Delta P}{P} + 2\frac{\Delta d}{d} + 4\frac{\Delta T}{T}$
- $\frac{\Delta e}{e} = \frac{0.1}{3.0} + \frac{2 \times 0.01}{6.14} + \frac{4 \times 1}{500} = 0.04459$
- $\Delta e = 0.04459 \times e = 0.04459 \times 0.2859$
- Δe = 0.01275 = 0.01 (1 s.f.) *[1 mark]*

State the final answer to an appropriate degree of precision:

- e = 0.29 ± 0.01 *[1 mark]*

[Total: 3 marks]

To propagate fractional uncertainties with and without quantities raised to a power:

$$\text{If } y = \frac{ab}{c} \text{ then } \frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$$

$$\text{If } y = a^n b^m \text{ then } \frac{\Delta y}{y} = n\frac{\Delta a}{a} + m\frac{\Delta b}{b}$$

This is why the fractional uncertainty in d^2 is $2 \times \frac{\Delta d}{d}$ and the fractional uncertainty in T^4 is $4 \times \frac{\Delta T}{T}$

When quoting values with their uncertainties, an 'appropriate degree of precision' means the uncertainty should not have a greater level of precision than the value

If e was quoted to 3 s.f., an appropriate degree of precision would be e = 0.286 ± 0.013

Question 11 Hard [16 marks]

Part (a) [5 marks]

Fig. 1.1 shows a circuit designed by a student investigating the variation of potential difference with time as a capacitor charges and discharges whilst maintaining a constant current.

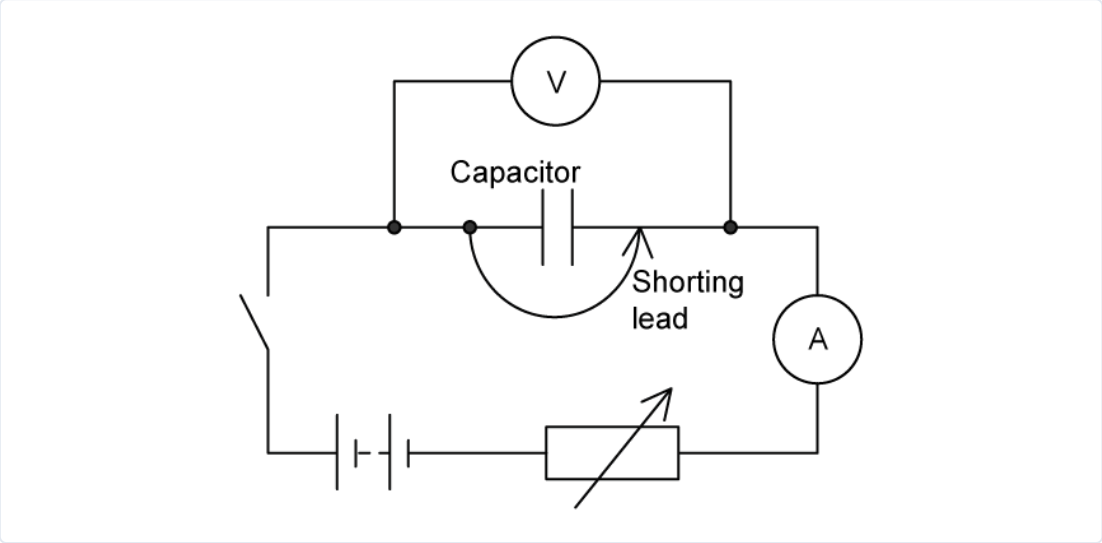


Fig. 1.1

The power supply has an e.m.f. of 6.0 V and negligible internal resistance. During the charging process, the maximum value of the resistance of the variable resistor is 100 kΩ.

(i) Suggest **one** advantage of using an analogue ammeter rather than a digital ammeter for this experiment.

[1]

(ii) Fig. 1.2 shows three analogue ammeters which can detect currents up to a maximum of 50 μA, 100 μA or 500 μA. Suggest, with a calculation, which of the analogue ammeters in Fig. 1.2 would be most suitable for the experiment.



Fig. 1.2

[3]

(iii) State the uncertainty in a reading of current using the ammeter you choose in **(ii)**.

[1]

Mark Scheme (5 marks):

(i) An advantage of using an analogue ammeter is:

- The position on scale can be marked (as it needs to be kept constant) / easier to read quickly / easier to detect changes; *[1 mark]*

(ii) The most suitable ammeter for this experiment is the:

- 100 μA ammeter; *[1 mark]*

Calculation:

- Maximum resistance, R = 100 kΩ = 100 × 10^3 Ω
- e.m.f. of the supply, V = 6.0 V
- Initial current: $V = IR \Rightarrow I = \frac{V}{R}$
- Initial current: $I = \frac{6.0}{100 \times 10^3}$ = 6 × 10^{-5} A = 60 μA *[1 mark]*

Conclusion:

*Any **one** from:*

- 100 μA is high enough to detect the 60 μA current and is a suitable for detecting changes; *[1 mark]*
- 50 μA is too low to detect the 60 μA current; *[1 mark]*
- 500 μA might not be sensitive enough to detect readings / changes in readings; *[1 mark]*

(iii) Uncertainty in 100 μA ammeter readings:

- Smallest division = 2 μA
- Uncertainty in a reading = ± ½ × smallest division = ±(½ × 2)
- Uncertainty in a reading = ± 1 μA *[1 mark]*

[Total: 5 marks]

Part (b) **[4 marks]**

A digital voltmeter with a three-digit display is used to measure the potential difference across the capacitor. A reading made during the charging process is shown in Fig. 1.3.



Fig. 1.3

The manufacturers of the meter state that its accuracy is ±2% and ±1 digit.

(i) Determine the maximum possible value of the potential difference across the capacitor at the moment the reading in Fig. 1.3 was taken.

[2]

(ii) The reading on the voltmeter has high precision. State and explain why the reading may not be accurate.

[2]

Mark Scheme (4 marks):

(i) Determine the maximum possible value of V:

- Absolute uncertainty in V = 2% of 2.23 V
- Absolute uncertainty in $V = \frac{2}{100} \times 2.23 = \pm 0.04$ V *[1 mark]*
- Max. value of V = 2.23 + 0.04 + 0.01 = 2.28 V *[1 mark]*

(ii) The reading may not be accurate because:

- There may be a zero error / calibration error / systematic error; *[1 mark]*
- Which makes all readings either higher or lower than true value; *[1 mark]*

[Total: 4 marks]

*Don't forget to add the ±0.01 in part **(i)**!*

Where the question says the meter has an accuracy of ±1 digit, this refers to the uncertainty in the final digit of the three-digit display on top of the uncertainty of 2%, hence the total uncertainty is 0.04 + 0.01 = ±0.05 V

Part (c) [3 marks]

Initially, the shorting lead is connected across the capacitor, as shown in Fig. 1.1, to fully discharge it. The timer is started at the moment the shorting lead is disconnected as the capacitor begins to charge.

The student measures the potential difference across the capacitor at 5 s intervals while adjusting the variable resistor to keep the charging current constant.

The label on the capacitor states its capacitance C is 510 μF.

Determine the number of different readings the student will be able to take before the capacitor becomes fully charged.

The charge on the capacitor is equal to $Q = CV$

Mark Scheme (3 marks):

Determine the number of different readings the student will be able to take:

List the known quantities:

- Capacitance, C = 510 μF = 510 × 10^{-6} F
- Potential difference when fully charged, V = 6.0 V
- Current, I = 60 μA = 60 × 10^{-6} A

Determine the total charge on the capacitor when fully charged:

- Total charge: $Q = CV = 510 \times 10^{-6} \times 6.0$
- Q = 3.06 × 10^{-3} C = 3.06 mC *[1 mark]*

Determine the time taken for the capacitor to fully charge:

- Current: $I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I}$
- Time: $t = \frac{3.06 \times 10^{-3}}{60 \times 10^{-6}}$ = 51 s *[1 mark]*

If a reading is taken every 5 s...

- Number of readings = $\frac{51}{5}$ = 10.2
- Number of readings = 10 *[1 mark]*

[Total: 3 marks]

Part (d) **[4 marks]**

The capacitor used in the experiment has a manufacturing tolerance of ±10%. Throughout the experiment, the charging current is maintained at the value calculated in **(a)**.

The data from the experiment produces a straight-line graph for the variation of potential difference with time. This shows that the potential difference across the capacitor increases at a rate of 110 mV s^{-1}.

(i) Calculate the true capacitance of the capacitor.

[2]

(ii) Deduce whether the capacitor is within the manufacturer's tolerance.

[2]

Mark Scheme (4 marks):

List the known quantities:

- Charging current, I = 60 μA = 60 × 10^{-6} A
- Rate of p.d. increase, $\frac{V}{t}$ = 110 mV s^{-1} = 110 × 10^{-3} V s^{-1}
- Tolerance / uncertainty in capacitance = ±10%
- Quoted value = 510 μF

(i) Calculate the true capacitance of the capacitor:

- Total charge: $I = \frac{Q}{t} \Rightarrow Q = It = 60 \times 10^{-6} \times t$
- Rate of p.d. increase = $\frac{V}{t} \Rightarrow V = 110 \times 10^{-3} \times t$
- Capacitance: $Q = CV \Rightarrow C = \frac{Q}{V}$
- $C = \frac{60 \times 10^{-6} \times t}{110 \times 10^{-3} \times t}$ *[1 mark]*
- C = 5.45 × 10^{-4} F = 545 μF *[1 mark]*

(ii) Deduce whether the capacitor is within the manufacturer's tolerance:

EITHER

Method 1: Determine the uncertainty in C

- Absolute uncertainty in $C = 510 \times \frac{10}{100}$ = ± 51 μF *[1 mark]*
- The capacitor is within the manufacturer's tolerance because 545 μF lies within the range 459 – 561 μF; *[1 mark]*

OR

Method 2: Determine the percentage difference

- Percentage difference = $\frac{545 - 510}{510} \times 100\%$ = 6.9%; *[1 mark]*
- The % difference between the values of capacitance is 6.9% which is within 10%; *[1 mark]*

[Total: 4 marks]

Question 12 Hard [16 marks]

Part (a) [2 marks]

An object of mass 50.00 g is placed on four different balances. For each balance, the reading is taken five times, as shown in Table 1.1.

Table 1.1

Balance	Mass Reading / g				
	1st	2nd	3rd	4th	5th
1	50.00	50.00	50.02	50.01	50.02
2	50.12	49.94	50.04	49.89	49.95
3	50.52	50.53	50.52	50.54	50.54
4	48.9	49.6	50.2	50.9	51.1

Analyse the data and conclude which sets of readings are accurate, precise or neither.

Mark Scheme (2 marks):

Analysis of the data:

- Take a mean of each set of readings; *[1 mark]*
- Write a conclusion about each set of readings using the words accurate, precise, inaccurate and imprecise with correct reasoning; *[1 mark]*

Example of analysis:

- Balance 1 mean = 50.01 g
- Balance 2 mean = 49.99 g
- Balance 3 mean = 50.53 g
- Balance 4 mean = 50.1 g
- Balances 1 & 2 show accurate readings
- Because their means are close to the true value
- Balances 1 & 3 show precise readings
- Because all readings are close to their mean value

- Balances 3 & 4 show inaccurate readings
- Because their mean values are not as close to the true value as the 1 & 2
- Balances 2 & 4 show imprecise readings
- Because the readings are widely scattered about their mean value
- Because balance 4 measures to a lower level of precision / number of decimal places than the other three

[Total: 2 marks]

Part (b) **[4 marks]**

A titanium sphere and a titanium cylinder were used in an experiment to measure the density of titanium. The diameter of the sphere was measured using a micrometer, and the diameter of the cylinder was measured using vernier callipers, as shown in Fig. 1.1.

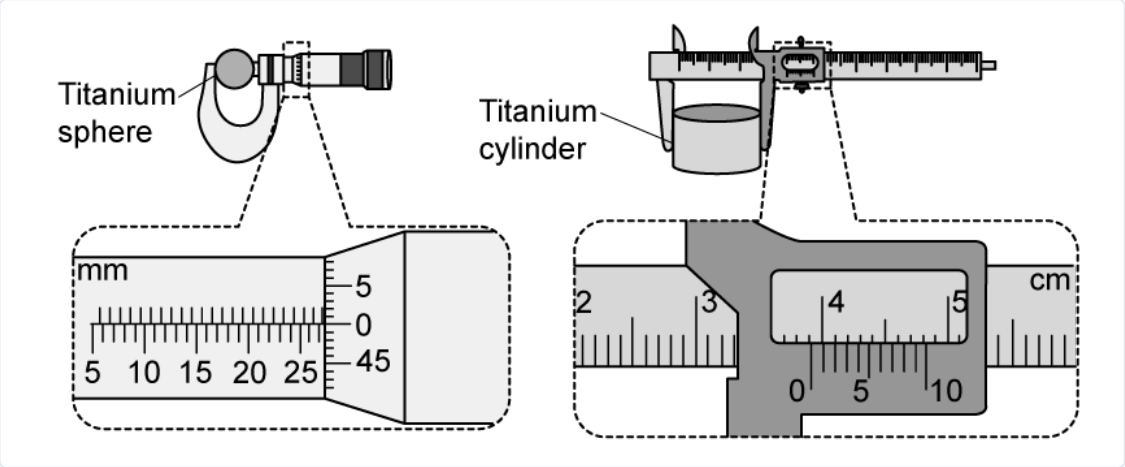


Fig. 1.1

Using Fig. 1.1, determine

(i) the diameter of the titanium sphere, in cm, with its uncertainty.

[2]

(ii) the diameter of the titanium cylinder, in cm, with its uncertainty.

[2]

Mark Scheme (4 marks):

(i) Determine the diameter of the titanium sphere with its uncertainty:

Calculate the diameter of the sphere in cm:

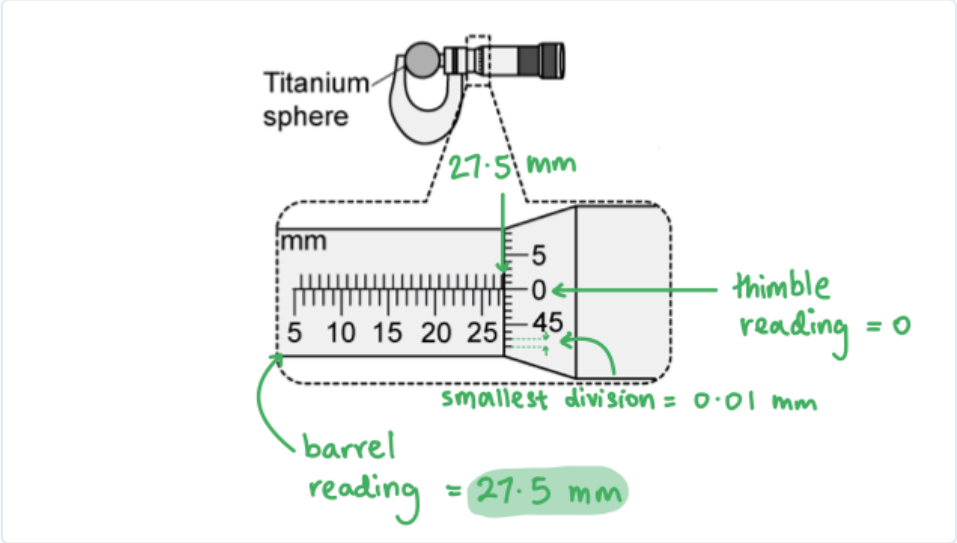
- From Fig. 1.1:
  - Reading on the barrel = 27.5 mm

- Reading on the thimble = 0
- Final reading = 27.50 mm

- Diameter of the sphere = 2.750 cm *[1 mark]*

Calculate the uncertainty in the diameter of the sphere:

- Uncertainty in the reading = ± smallest division
- Uncertainty in the reading = ± 0.01 mm = ± 0.001 cm *[1 mark]*
- Therefore, diameter of the sphere = 2.750 ± 0.001 cm



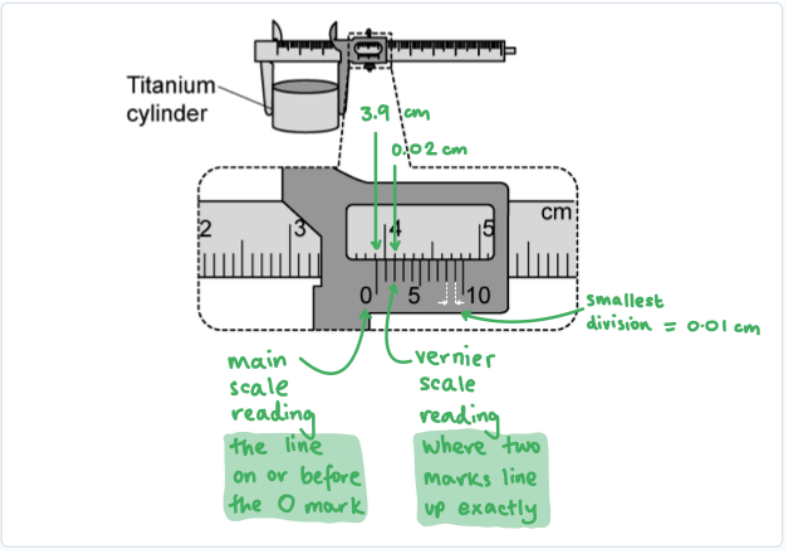
(ii) Determine the diameter of the titanium cylinder with its uncertainty:

Calculate the diameter of the cylinder in cm:

- From Fig. 1.1:
  - Reading on the main scale = 3.9 cm
  - Reading on the vernier scale = 0.02 cm
  - Final reading = 3.92 cm
- Diameter of the cylinder = 3.92 cm *[1 mark]*

Calculate the uncertainty in the diameter of the cylinder:

- Uncertainty in the reading = ± smallest division
- Uncertainty in the reading = ± 0.01 cm *[1 mark]*
- Therefore, diameter of the cylinder = 3.92 ± 0.01 cm



[Total: 4 marks]

Part (c) [6 marks]

The measurements for the mass and height of the cylinder with their fractional uncertainties are shown in Table 1.2.

Table 1.2

	measurement	**fractional uncertainty**
cylinder mass / g	774.8	0.03
cylinder height / cm	15.0	0.01

The object used in **(a)** was the titanium sphere. The mass of the sphere was measured using the balance 2 shown in Table 1.1.

Determine

(i) the density of the titanium sphere, in g cm^{-3}, and its associated percentage uncertainty.

[3]

(ii) the density of the titanium cylinder, in g cm^{-3}, and its associated percentage uncertainty.

[3]

Mark Scheme (6 marks):

(i) Determine the density of the titanium sphere:

List the known quantities:

- Sphere diameter, d = 2.750 ± 0.001 cm (from **(b)**)
- Mean sphere mass, m = 49.99 g (from **(a)**)

Calculate the density of the sphere:

- Volume of a sphere: $V = \frac{4}{3}\pi r^3 = \frac{4\pi}{3}\frac{d}{2}^3$
- Density: $\rho = \frac{m}{V} = \frac{3m}{4\pi}\frac{2}{d}^3$
- Density: $\rho = \frac{3\times 49.99}{4\pi}\frac{2}{2.750}^3$
- Density: ρ = 4.5908 = 4.59 g cm^{-3} *[1 mark]*

Convert the uncertainties into percentage uncertainties:

- Uncertainty in mass, Δm = (½ × range) = ½ × (50.12 – 49.89) = ±0.115 g *[1 mark]*
- % uncertainty in m: $\%\Delta m = \frac{0.115}{49.99}\times 100\% = 0.23\%$
- % uncertainty in d: $\%\Delta d = \frac{0.001}{2.750}\times 100\% = 0.036\%$

Calculate the percentage uncertainty in density:

- $\%\Delta\rho = \%\Delta m + 3\%\Delta d$
- $\%\Delta\rho$ = 0.23 + (3 × 0.036) = 0.34% *[1 mark]*

Quote the density with its associated percentage uncertainty:

- Density of titanium sphere, ρ = 4.59 g cm^{-3} ± 0.34%

(ii) Determine the density of the titanium cylinder:

List the known quantities:

- Cylinder diameter, d = 3.92 ± 0.01 cm (from (b))
- Cylinder mass, m = 774.8 g
- Cylinder height, h = 15.0 cm
- Fractional uncertainty in mass, $\Delta m/m$ = 0.03
- Fractional uncertainty in height, $\Delta h/h$ = 0.01

Calculate the density of the cylinder:

- Volume of a cylinder: $V = \pi r^2 h = \pi h\frac{d}{2}^2$
- Density: $\rho = \frac{m}{V} = \frac{m}{\pi h}\frac{2}{d}^2$
- Density: $\rho = \frac{774.8}{\pi\times 15.0}\frac{2}{3.92}^2$ *[1 mark]*
- Density: ρ = 4.2799 = 4.3 g cm^{-3} *[1 mark]*

Calculate the percentage uncertainty in density:

- $\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta h}{h} + 2\frac{\Delta d}{d}$
- $\frac{\Delta\rho}{\rho} = 0.03 + 0.01 + 2\frac{0.01}{3.92} = 0.0451$
- % uncertainty in $\rho = 4.5\%$ *[1 mark]*

Quote the density with its associated percentage uncertainty:

- Density of titanium cylinder = 4.3 g cm^{-3} ± 4.5%

[Total: 6 marks]

Remember, if a quantity has a power associated with it then you must multiply the propagated uncertainty by that power. This is because a quantity with a power has a larger effect on the calculated quantity, hence its fractional uncertainty will be higher.

In the case of the cylinder, the diameter is squared, hence its fractional, or percentage, uncertainty is twice as large since the measured diameter has such a great effect on the final value of ρ.

In the case of the sphere, the diameter is cubed which has an even larger effect on the final value of ρ as the fractional uncertainty must be multiplied by 3.

Part (d) **[4 marks]**

The true value of the density of the titanium used in both the cylinder and the sphere was found to be 4.51 g cm^{-3}.

The student assumes that the percentage difference between the density of the titanium cylinder and the true value is solely due to the uncertainty in the volume and the uncertainty in the mass.

Assess the validity of this assumption.

Mark Scheme (4 marks):

List the known quantities:

- True value of density = 4.51 g cm^{-3}
- Sphere density = 4.59 g cm^{-3}
- Cylinder density = 4.3 g cm^{-3}
- % uncertainty in sphere density = 0.3%
- % uncertainty in cylinder density = 4.5%

Calculate the percentage difference from the sphere's density and the true value:

- % difference (sphere) = $\frac{4.59 - 4.51}{4.51} \times 100 = 1.8\%$

Calculate the percentage difference from the cylinder's density and the true value:

- % difference (cylinder) = $\frac{4.51 - 4.3}{4.51} \times 100 = 4.7\%$

Compare the percentage difference with the calculated uncertainty for the sphere:

- Sphere: 1.8 – 0.3 = 1.5% *[1 mark]*
- There is a significant error (>1%) unaccounted for by the mass or volume, hence the assumption may not be valid for the sphere; *[1 mark]*

Compare the percentage difference with the calculated uncertainty for the cylinder:

- Cylinder: 4.7 – 4.5 = 0.2% *[1 mark]*
- There is only a small error (<1%) unaccounted for by the mass or volume, hence the assumption can be said to be valid for the cylinder; *[1 mark]*

[Total: 4 marks]

Always read the question carefully, in this case, you are trying to find out if the student's assumption is correct.

*According to the density equation ($\rho = m/V$), ρ **only** depends on m and V, so the percentage difference and the percentage uncertainties calculated in **(c)** should be the same if the assumption is correct. In other words, the difference between the percentage difference with the true value and the uncertainties calculated in **(c)** should be close to zero for the assumption to be valid.*

You may have automatically assumed that this assumption must be correct, but working through the maths shows it isn't. Measurement is a science in itself, and there are many ways that errors and uncertainties creep in, such as random errors or unknowns such as impurities in the metal